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ASSOCIATED DETAIL SPECIFICATION

MANUAL, OPERATIONAL STATIONS BOOK (OSB) REQUIREMENTS FOR DESIGN

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification sets forth requirements for a Design Operational Stations Book (OSB), employed to describe how the ship's personnel operate within the shipboard operational systems, for the establishment of team training of military personnel (not individual operator training) within described operational spaces and stations and as a historical record of the design philosophy intended by the designer, or the effect of the design upon operations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 03R42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N6946 AREA TMSS
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

SPECIFICATIONS

MILITARY

MIL-M-24784 - Manual, Technical: General Acquisition and Development Requirements.
MIL-M-24784/1 - Manual, Technical Change Package Requirements.
MIL-M-24784/2 - Manual, Technical Revision Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

DEPARTMENT OF DEFENSE (DoD)

DOD 5200.1-R - Information Security Program Regulation.
DOD 5220.22-M - Industrial Security Manual for Safeguarding Classified Information.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF THE NAVY

0900-LP-005-1010 - Technical Manual for Design Work Study.

(Application for copies should be addressed to the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

U.S. GOVERNMENT PRINTING OFFICE

Z253.U58 - US Government Printing Office Style Manual.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications), the text of this specification shall take precedence. When a generic requirement is in conflict with a unique requirement from the governing associated detail specification, the unique requirement shall take precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. In order to develop and maintain the greatest effectiveness and efficiency in the design of new shipboard integrated combat systems, the improvement (modernization) of older systems, and the development of their associated documentation requirements, work study procedures employ analytical

shall persist until the

techniques to establish and optimize relationships between components of the shipboard operational system - human factors, equipment factors and interfacing software support requirements, and operational concepts and environments. The OSB shall describe these relationships with particular attention to:

(a) The operational (rather than the technical) system.
(b) The man, machine, software interface.
(c) Optimum system employment as developed through analytical techniques.
(d) Operating and system set-up procedures for all modes of operation, including casualty conditions and special evolutions applicable to the system.
(e) Rationale, including work study, to support the system design concept and the developed operational procedures.
(f) Sequenced, step-by-step, operator procedures, and operator actions or interactions (see figures 5-1 and 5-2).
(g) Manning requirements for conditions I and III, the governing readiness condition (if other than conditions I and III), special sea detail, and predominate special evolutions (for example, underway replenishment, vertical replenishment, and so forth) (see figure 5-3).
(h) Special skills, knowledge, or training required to operate the system.
(i) Providing a source of information not readily available from other sources.

3.1.1 Space applicability. When specified (see 6.2), the OSB shall define and describe the relationships of shipboard operational systems and system components for the spaces. The size, mission and tasks of the ship shall be used to determine the shipboard spaces, stations, and systems to be included in the OSB. Spaces and stations specified as follows are typical of those which may be considered for inclusion in the OSB:

(a) Pilot house and bridge wings (including exposed and auxiliary conning stations).
(b) Chart room.
(c) Signal station and signal bridge.
(d) Secondary conning station.
(e) Weapons control stations.
(f) Flag bridge and command and control complex (including tactical flag command center).
(g) Combat information center complex (including sonar control room, antisubmarine warfare module and central computer complex).
(h) Meteorological spaces and stations.
(i) Operational intelligence spaces.
(j) Communications complex.
(k) Engineering and machinery monitoring and control stations.
(l) Damage control stations.
(m) Spaces in which special evolutions are carried out, such as:

    (1) Aircraft operations control (fixed and rotary wing).
    (2) Aircraft handling and maintenance control.
    (3) Amphibious and landing force command and control.

functions responsible for assigning ASR's within the

(4) Ammunition and cargo handling.
(5) Squadron ready rooms.
(6) Deck operations (such as handling of mine countermeasures equipment, replenishment, beaching equipment, rescue and salvage equipments).

3.2 Technical accuracy. Information in the manual shall accurately represent the system, subsystem, equipment or function being described. Technical inaccuracies or inconsistencies shall be considered justification for rejecting a technical manual.

3.3 Ensurance of complete and up-to-date data. Information contained in an OSB shall be checked against established Chief of Naval Operations, Department of the Navy (CNO) manning authorizations, actual shipboard equipment lists (such as drawings, specifications, ship characteristics and mock-ups), latest shipboard arrangement configurations, and documented information sources, immediately prior to submission for review and approval.

3.4 Preparation. Unless otherwise specified herein, the OSB shall be in accordance with the data ordering documents and the requirements specified herein. The OSB shall be prepared by personnel who are:

(a) Knowledgeable in shipboard operational procedures.
(b) Knowledgeable in design work study techniques.
(c) Knowledgeable in interior and exterior communications systems.
(d) Knowledgeable in software support of integrated combat systems (as applicable to the system being described).

3.4.1 Preparation techniques. The OSB shall be in consonance with the application of design work study techniques employed in the early stages of design, to the degree specified by the Government, to ensure functional and operational accuracy. Where the application of design work study techniques is not specified by the Government in the design stages, an adequate level of design work study, as specified in flow process charts, as a minimum (see 3.1(f) and 3.10.3.2.2.2). Primary applicable design work study techniques are:

(a) Functional analysis.
(b) Functional flow diagrams (FFD).
(c) Operational sequence diagrams (OSD).

3.4.2 Layout. The OSB layout shall provide for ease of use so that it may be directly promulgated as ship's doctrine. The OSB shall be in loose leaf form so that parts may be separated from the basic publication for use by individual operators.

3.5 Security classifications. The overall classification assigned to a technical manual shall agree with the highest classification assigned to any portion within and shall be placed conspicuously at the top and bottom of the Cover and Title page or abbreviated title. Security classification markings shall be in accordance with DOD 5220.22-M and 5200.1-R and shall be placed in accordance with DOD 5220.22-M and 5200.1-R. The security markings shall be the same as for other title pages.

3.5.1 Classified information. Wherever possible, the manual shall be void of classified material. Information classified higher than confidential shall not be included in the manual. Confidential material shall be kept to a minimum and, if possible, shall be assembled in an appendix or other division of the manual, for ease of identification, removal, and separate distribution where desired. In addition to the classified security markings, each page which is classified shall have each paragraph on that page identified by its classification by use of a capitalized English letter indicating the classification, enclosed in parentheses: example, (U) or (C). If the manual is classified, the list of effective pages shall identify the pages which are classified.

3.6 Deliverable products and data items. The deliverable products and data items shall be specified in the Contract Data Requirements List (CDRL) and shall conform to the requirements specified in MIL-M-24784 and herein. Data items shall include the following, as applicable:

(a) Outline or book plan.
(b) Review draft copy (RDC).
(c) Preliminary technical manual (PTM).
(d) Final reproducible copy (FRC).
(e) Supplements.
(f) Changes (see 3.6.1).
(g) Revisions (see 3.6.2).
(h) Management data (see 3.6.3).

3.6.1 Change packages. When specified (see 6.2), change packages shall be in accordance with MIL-M-24784/1.

3.6.2 Revisions. When specified (see 6.2), revisions shall be in accordance with MIL-M-24784/2.

3.6.3 Management data. Unless otherwise specified (see 6.2), the following quality assurance data items shall be provided in accordance with MIL-M-24784:

(a) Technical Manual Quality Assurance Program Plan.
(b) Validation Plan.
(c) Validation Certification.
(d) Technical Manual Evaluation Records.
(e) Verification Plan.
(f) Technical Manual Verification Incorporation Certification.

3.7 Arrangement. Unless otherwise specified, the manual shall be arranged in a standardized format (that is, front matter, technical content, appendices, glossaries, indexes and back matter) and appropriately divided by volume, part, chapter and section in accordance with the requirements of MIL-M-24784, Appendix C and the following.

3.7.1 Cover and title page and outline prime title. The prime title for a Design OSB Outline and a Design OSB Manual shall be as follows:

OPERATIONAL STATIONS BOOK (DESIGN)
FOR
(INDICATE SPECIFIC SHIP AND CLASS)
(VOLUME NO., IF MULTIPLE VOLUMES)
(OPERATIONAL AREA(S) OR SYSTEM DESCRIBED)

3.7.2 Introduction. The OSB introduction shall state the purpose of the OSB, and its intended use. The introduction shall, as a minimum, include the following:

(a) A short explanation of the design philosophy used to develop the operational system.
(b) A brief description of the design criteria that optimized the procedures and minimized the manning requirements within the configuration.
(c) A statement indicating that the procedures were developed within the arrangement according to analytical techniques, and that any changes in the arrangement might severely hamper the effectiveness and efficiency of the system.
(d) An explanation as to why this configuration, with the operational procedures contained in the OSB, benefits the operator. A statement shall be included to the effect that the operator was given prime consideration throughout the development of the procedures presented.
(e) A statement of reporting errors (see MIL-M-24784).

3.8 Format and preparation instructions.

3.8.1 Writing style. Unless otherwise specified, the style of writing, level of writing, readability, and referencing shall be in accordance with the requirements of MIL-M-24784, Appendix B.

3.8.2 Safety precautions. Unless otherwise specified, safety and health warnings, cautions and notes shall be in accordance with the requirements of MIL-M-24784, Appendix D.

3.8.3 Tabular material. Unless otherwise specified, the preparation of tables shall be in accordance with the requirements of MIL-M-24784, Appendix E.

3.8.4 Graphics. Unless otherwise specified, the preparation of illustrations, drawings, diagrams and sketches shall be in accordance with the requirements of MIL-M-24784, Appendix F.

3.8.5 Numbering. Unless otherwise specified, the numbering of paragraphs, procedures, divisions, issues, changes, illustrations, tables, and publications shall be in accordance with the requirements of MIL-M-24784, Appendix G.

3.9 Contents. The contents shall be limited to information required by the user (operator) to understand and carry out the required functions of the shipboard operational system. Only major functional units of equipment shall be included in the OSB. The contents shall be organized in a form so that only those changes which affect the ships functional operations will require revision to the OSB. To avoid the requirement for minor revisions, the following shall not be included in the OSB:

(a) Specific military designations for equipment or system nomenclatures (use of general terms is intended).
(b) Course numbers for service schools or correspondence courses.
(c) Specific characteristics of the ship, systems, or equipment.
(d) Listings of items such as equipment, systems, or armaments.
(e) Equipment operating instructions contained in technical manuals.

3.9.1 Symbology. The symbology employed on the contents of the OSB shall be standardized to those used in accordance with NAVSEA 0900-LP-005-1010. It is extremely important that the OSB users have a thorough understanding of the meaning and use of the various symbols. Symbols not only provide the design engineer a means of examining the numerous operational tasks to be performed, but once established, if properly interpreted and employed, the symbols greatly facilitate and simplify operator training and user understanding of the functions to be performed. The symbols to be employed in the OSB to denote actions shall be as specified in 3.9.1.1 through 3.9.1.7 and sequential employment of symbology shall be as shown on figures 5-1 and 5-2.

3.9.1.1 Transport. An open arrow shall be employed to denote movement of personnel material, or transfer of information (see figure 5-1).

3.9.1.2 Operation. An open circle shall be employed to denote an operation performed (see figure 5-1).

3.9.1.3 Inspection. An open rectangle shall be employed to denote a test or examination (see figure 5-1).

3.9.1.4 Mandatory actions. Solid line symbols shall be employed to denote mandatory actions. Solid lines connecting the symbols shall be employed to denote mandatory sequential actions (see figures 5-1 and 5-2).

3.9.1.5 Optional actions. Dash (broken line) symbols shall be employed to denote optional actions. Dash lines connecting symbols shall be employed to denote optional sequential actions (see figures 5-1 and 5-2).

3.9.1.6 Double Symbols. Combined symbols shall be employed where the exterior symbol denotes the principal action and the interior symbol denotes the secondary action within the sequence of events (see figure 5-2).

3.9.1.7 Symbol identification. If identified, symbols for primary actions shall be sequentially identified for each type of action by means of numerically increasing Arabic numerals placed in the center of the symbols. Where symbols representing secondary actions are employed and identified, they shall be sequentially identified for each type of action alphabetically by means of small English letters placed in the center of the symbols (see figure 5-2).

3.10 Text development. The OSB shall contain only essential information and shall be factual, specific, concise, comprehensive, and unambiguous. It shall be so clearly worded as to be readily understandable at levels specified in MIL-M-24784, Appendix B. The U.S. Government Printing Office Style Manual shall be used as a general guide for capitalization, punctuation, compound word forms, numerals in text, and spelling. The OSB shall be developed with particular attention to the requirements of 3.1 and 3.9.

3.10.1 Development guidance. Sample OSB for guidance and technical direction in preparation of an OSB shall be provided by the Government upon request.

3.10.2 Design OSB. A OSB shall normally be prepared during the later stages of concept design or early in the preliminary design stage. It shall be developed prior to detailed design and shall be used for:

(a) Guidance and development of a subsequent Construction and Major Modification OSB.
(b) Computer programming.
(c) Manning and training determinations.
(d) Ship design modifications.
(e) Operational forces comments.
(f) Design approval documentation.

3.10.3 Informative chapters. The first page of each chapter shall provide a list of sections and the primary articles therein listed with titles and page numbers.

3.10.3.1 Chapter 1 (general information). If the OSB covers several major systems, this chapter shall include general information on the overall operational concepts of the ship, to include:

(a) Missions and tasks (if this can be published as unclassified information).
(b) General characteristics which were developed by design work study techniques in the design layout of the ship and which are not covered in subsequent chapters, such as the relationship of areas to each other.
(c) Operational concepts of the ship indicating the major functions to be performed and the interrelationship of the areas to be covered in subsequent chapters.

3.10.3.2 Other chapters. Each chapter following Chapter 1 shall describe a specific physical area (space or station) on the ship. Modularized (functional) areas within a space shall be presented in sections of the chapter. For those spaces which are not described by use of functional areas, all information shall be described in the chapter without section numbering. The requirements of 3.10.3.2.2.1 shall be presented in the general articles and the requirements of 3.10.3.2.2.2 shall be presented in the other articles of this unsectionalized chapter.

3.10.3.2.1 Section I (general information). Section I shall be the general section and shall provide general information on its subject space. The articles contained in this chapter shall be numbered: -101 through -199 (dashes denote the number of the appropriate chapter). As a minimum, the general section shall include:

(a) General functions within the overall space.
(b) Operation and supervision of the space, describing any variations for special evolutions or conditions, including emergencies and casualties.

whether or not time is transmitted to the command work.

(c) Team training required for the personnel manning of the space, in general terms unless specific requirements are furnished by the Government (in all cases specific course numbers shall be avoided).
(d) List of functional areas within the space (indicating the section of the chapter for the area described).
(e) Operational aids required in the area (for example, publication records, reports, and portable equipment).
(f) Diagram of the space showing areas (labeling functional areas by name and referencing the applicable OSB section number) (see figure 5-4).
(g) Where applicable, a matrix indicating which areas are manned under each operational condition or evolution.
(h) Key plan of the space showing its location within the ship (see figure 5-5).

3.10.3.2.2 Other sections (describing modular areas).

3.10.3.2.2.1 General articles. General articles shall provide general information on the modular (functional) area described by the parent section. General articles shall be numbered: --01 through --09 (dashes denote the number of the appropriate chapter and section). As a minimum the general articles shall describe the following:

(a) General functions within the area.
(b) Operation and supervision of the area, describing any variations for special evolutions or conditions, including emergencies and casualties.
(c) Team training required for personnel manning the module, in general terms unless specific requirements are furnished by the Government.
(d) List of operational positions within the area (indicating the article in which the operational position is described).
(e) Operational aids required in the area (for example, publication records, reports, and portable equipment).
(f) Diagram of the area showing the following:

(1) Ship centerline with forward arrow, if applicable, and major frame numbers.
(2) Major equipment.
(3) Positions manned during evolutions with maximum manning (labeled by the article describing the position).
(4) Tabulation of station article numbers with corresponding operational position titles for the evolution depicted.
(5) If feasible, within areas with repetitive operations (for example, communications space), major flow of information indicated by a line with an arrow (labeled with the method of transmission and circuit, if applicable).

(g) Manning matrix indicating which positions (by position name and article number versus evolutions) are manned during each evolution. As a minimum manning for readiness conditions I and III, special sea detail, and the controlling special evolution (for example, flight quarters, underway replenishment, and so forth), as applicable to the space or area depicted shall be listed. An "X" shall be used to indicate that the position is manned, a "-" shall be used to indicate that the position is not manned, an "0" shall be used to indicate optional manning, and "W/R" shall be used to indicate manned when required (see figure 5-3).

3.10.3.2.2.2 Other articles. Other articles shall describe specific operational positions. These articles shall be numbered: --10 through --99 (dashes denote the number of the specific chapter and section). As a minimum these articles shall describe the following:

(a) Name of operational position.
(b) Station and equipment manned.
(c) When manned.
(d) General training requirements for manning the position or equipment shall be indicated unless specific requirements are provided by the Government, in which case the specific requirements shall be described. Training requirements shall include:

(1) Schools or course descriptions (not titles or numbers).
(2) Reference publications for study. Naval Warfare Publications (NWP) and similar authoritative publications may be included, but reference to specific manuals or instructions, by the exact title or number shall be avoided.
(3) If applicable, qualifications in other positions or jobs. If developed and provided by the command or agency concerned, qualification standards and or special skills for the position shall be included in this part of the text or as an appendix, whichever is appropriate.

(e) Facilities (for example, communications and/or special tools or equipment) required in the position for carrying out duties. Where a selection of facilities is available, the facilities may either be referenced to those listed for the module or those for other positions in the module.
(f) Duties and responsibilities of the operational position, including a description of the operating procedures for the position, any variations for casualties, special evolutions or conditions. Command and control spaces, at least one alternate position (method) shall be indicated for a major casualty to the equipment at the position described. The description shall be thorough enough to cover the operational concepts, preparation of the equipment (tuning and watch standing routines), reports, and information sent and received including, where applicable, transmission methods (type of system and circuit nomenclature) and terminal equipment (handset, headset, keyset, speaker, and so forth). The description shall be simple, concise, readable, and

not repeat information which is readily available through other publications or the previous training of the individual. Specific operating instructions for equipment (covered in manuals), procedures for communicating (covered by previous training) and responsibilities of such positions as commanding officer or officer-of-the-day (covered in current regulations and directives) shall not be specified. Duties which are applicable to the operational concepts of the system shall be specified. The operations shall be shown in a logic flow process chart.

3.11 Appendices. Where applicable, appendices shall be provided to present summations to clarify or assist ship personnel, as follows and as specified (see 6.2):

(a) Interior communications including an operational station versus circuit matrix identifying interior communications circuits and facilities available at each operational station and/or position within spaces or stations covered by the OSB. The matrix shall reflect operator position, circuit designation (by circuit name and designation), E-Call and selector facilities, and interphone capabilities for sound powered telephone systems, Naval Tactical Data Systems (NTDS) interphone facilities (CK), ship service telephones, announcing systems (for example, 1MC, or 5MC), intercommunications systems (amplified voice) (for example, 21MC, 39MC, and so forth), voice tubes, pneumatic tube facilities, and Interior Voice Communication Systems (IVCS), or equal capabilities.

(b) Exterior communications including a matrix identifying remote exterior communications circuits and facilities available at each operational station or position within spaces or stations covered by the OSB. The matrix shall reflect operator position, circuit designation channel selection capability and whether the circuit is at a remote individual terminal, incorporated within an NTDS console or subconsole, or a capability within IVCS. Additionally, normal mode of operation (for example, guard or monitor) shall be indicated for that position located. The matrix shall indicate send or receive terminal locations for teletype equipment (for example, LINK 14) within Command and Control and Combat System spaces and stations.

(c) Interior and exterior communications NTDS console matrix reflecting interior (inclusive of amplified voice) and exterior communications capabilities at each NTDS console or subconsole within spaces or stations covered by the OSB, when extraction of such facilities from complex interior and exterior communications matrices will enhance readability.

(d) Check-off list for evolutions such as casualty control.

(e) Special tools (if a separate complete listing would be useful.

(f) Separation of classified information from the remainder of the text (see 3.5.1).

(g) Qualification standards if provided by the command or agency concerned and not part of the text (see 3.10.3.2.2.2(d)(3)).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality control. Unless otherwise specified (see 6.2), the quality conformance inspection, Government inspection, validation, verification, technical manual quality assurance (TMQA) program reviews, technical manual evaluation records and rejection requirements shall be in accordance with MIL-M-24784.

5. PACKAGING

5.1 Packaging requirements. Unless otherwise specified (see 6.2), the packaging and preservation requirements for delivery of book plans, review draft copies, preliminary technical manuals, final reproducible copy, technical manuals, replenishment materials, changes and revisions shall be in accordance with MIL-M-24784.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Design OSB is intended to describe the ship's personnel operation within the shipboard operational systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification (or any TMCR referencing this specification).
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
(c) List of spaces to be included (see 3.1.1)
(d) Applicable publication number, classification, and distribution statement (see 3.1).

(e) An outline and book plan shall be specified whenever a technical manual is required for a new equipment or system (see 3.6a).
(f) Double column text shall normally be specified. When increased effectiveness of presentation results, a single column format is also acceptable.
(g) A review draft copy (RDC) shall always be specified for every new technical manual (see 3.6b). For change acquisition, unless changes to the technical manual have been pre-approved (TMDER, ECP, and so forth) by the Government, RDC shall be specified.
(h) Specify the number of weeks required for Government review of technical manual deliverables.
(i) Specify a preliminary technical manual when it is necessary to field equipment for test and evaluation, or when an extended period of use is required prior to Government verification (normally applies to complex equipment and systems only), see 3.6c.
(j) Final reproducible copy (see 3.6d).
(k) Supplement manuals (see 3.6e) shall be specified when:

(1) The classified information can be confined to the supplement manual such that its basic technical manual will be routed as "unclassified".
(2) It augments a technical manual to provide for a different model and can be justified as a cost effective method of promulgation.
(3) It can be justified as a practical method of promulgation.

(l) Changes shall be specified only when they pertain to technical manual corrections resulting from a hardware change or modification. They may also be specified if it has been determined that significant omissions of technical data or information has been identified. Do not specify a change to correct minor or editorial corrections. When changes are justified on the above basis, the change package shall include the correction of all outstanding temporary or interim changes and may include minor changes (see 3.6f). The change package shall be developed in accordance with MIL-M-24784/1.
(m) Whenever the number of change pages is estimated to exceed approximately 25 percent of the total number of pages in the manual, an update revision shall be specified (see 3.6g) and developed in accordance with MIL-M-24784/2.
(n) When the total number of pages requiring change exceed approximately 50 percent of the total number of pages in the manual, a complete revision shall be specified (see 3.6g) and developed in accordance with MIL-M-24784/2.
(o) What quality assurance and management data items are to be delivered (see 3.6.3, 4.2, 6.4 and MIL-M-24784).

(1) TMQA program plan.
(2) Validation plan.
(3) Validation certification.
(4) Evaluation records.

(5) Verification plan.
(6) Verification incorporation certification.

(p) Requirements for appendices, glossaries and indices (see 3.7).
(q) When preservation, packing, and marking are other than as specified (see Section 5. PACKAGING).
(u) For classified equipment, attach a DD Form 254 specifying the following:

(1) Security classification.
(2) Downgrading and declassification notification

this specification) must be listed on the Contract Data Requirements List (DD Form 1423) in order to acquire the technical manuals described by this specification, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

6.4 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title
3.6.3	DI-M-2194	Manual, Technical: Quality Assurance Program Plan
3.6.3	DI-M-2195	Manual, Technical: Validation Plan
3.6.3	DI-M-2196	Manual, Technical: Validation Certification
3.6.3	DI-M-2197	Manual, Technical: Evaluation Record
3.6.3	DI-M-2198	Manual, Technical: Verification Plan
3.6.3	DI-M-2201	Technical Manual Verification Incorporation Certification

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.5 Definitions. The words or phrases used throughout this specification are defined in MIL-M-24784.

6.6 Subject term (key word) listing.

Appendices
Cover document
Foreword
Record of changes
Symbology page
Text development

Review activity:
Navy - EC

Preparing activity:
Navy - SH
(Project TMSS-N242-05)

Supporting adaptive routing is the path quality matrix. This

SYMBOLOGY USAGE

Importance of symbols: The symbols emplyed in task descriptions contained in this book are those used in design work study technology. It is extremely important that OPERATORS, SUPERVISORS and all personnel who have occasion to use this book have a thorough understanding of the meaning and use of the various symbols. Symbols not only provide the analyst a means for examining the various tasks in the initial Design Work Study but more importantly, used and interpreted properly, greatly facilitiate and simplify OPERATOR training. Symbols employed in this book are as follows:



Transport. The movement of personnel, material or the transfer of information.



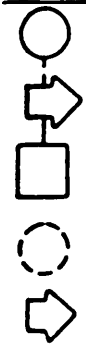
Operation. An operation performed.



Inspection. An examination or test.

FIGURE 5-1. Standard symbology.

System of notation:



Broken lines connecting symbols indicate an optional sequence. Action following a broken line need not be accomplished in the order it is placed in the text.

Solid lines connecting symbols indicate a mandatory sequence.

Broken line symbols indicate optional actions.

Solid line symbols indicate mandatory actions.

Reversed symbols indicate internal or external Transport



Internal Transport. Receipt of information or material from external source(s) or system(s).



External Transport. Generation of information or material to external source(s) or system(s).

Double symbols indicate combined actions:



An action implying first Inspection and second Operation. Exterior symbol denotes the principal action.

Example:



First primary Operation required to be performed. (Indicated by Arabic numerals sequentially within the symbols).

Second primary Operation required to be performed.

First Transport required to be performed.

First secondary Inspection. Lettered sequentially and indented out-of-line. Connected to [b] by broken line indicating that [a] and [b] need not be accomplished in order.

Second secondary Inspection.

Third primary Operation. (This is an example of symbology for an optional action).

Second primary Transport.

FIGURE 5-2. Symbology sequence system.

1 kHz digits. A frequency designator is normally used to

STATION MANNING MATRIX

LEGEND: X: MANNED -: NOT MANNED O: OPTIONAL MANNING W/R: MANNED WHEN REQUIRED		READINESS CONDITIONS			SPECIAL EVOLUTIONS		STATION REFERENCE		
OPERATIONAL STATIONS	STA ABBREV	CONDITION I	CONDITION	CONDITION III	SPECIAL SEA DETAIL		STATION ARTICLE	SMD STA. NO.	NOTES
PILOTHOUSE/BRIDGE									
Commanding Officer	CO	X	O	O	X		3110		
Navigator	NAV	W/R	W/R	W/R	X		3111		1
Officer of the Deck	OOD	X	X	X	X		3112		
Junior Officer of the Deck	JOOD	X	W/R	W/R	X		3113		
Assistant Navigator	ASTNAV	X	W/R	W/R	X		3114		2
Boatswain's Mate of the Watch	BMOW	X	-	-	X		3115		
Quartermaster of the Watch	QMOW	X	X	X	X		3116		
Ship Control Console Operator	SCCO	X	X	X	X		3117		
Messenger of the Watch/ Relief SCCO	MOW	X	W/R	X	X		3118		
SP Phone Talker (JA)	JA TLKR	X	W/R	-	-		3119		
Status Board Keeper/ Talker (JS)	BD KPR	X	X	-	X		3120		
BRIDGE WINGS (P/S)									
Bearing Taker (P/S)(JW)	BEARNG TKR	X	W/R	-	X		3121 3122		
Lookout (P/S)(JL)	LKOUT	X	X	X	X		3123 3124		
Lookout (Aft)(JL)	AFT LKOUT	X	X	X	X		3125		

NOTES:

1. Primary functions as TAO in CIC during evolutions requiring TAO.
2. Functions as bearing recorder (JW) during special sea detail evolutions.

FIGURE 5-3. Pilothouse and bridge wings manning requirements.

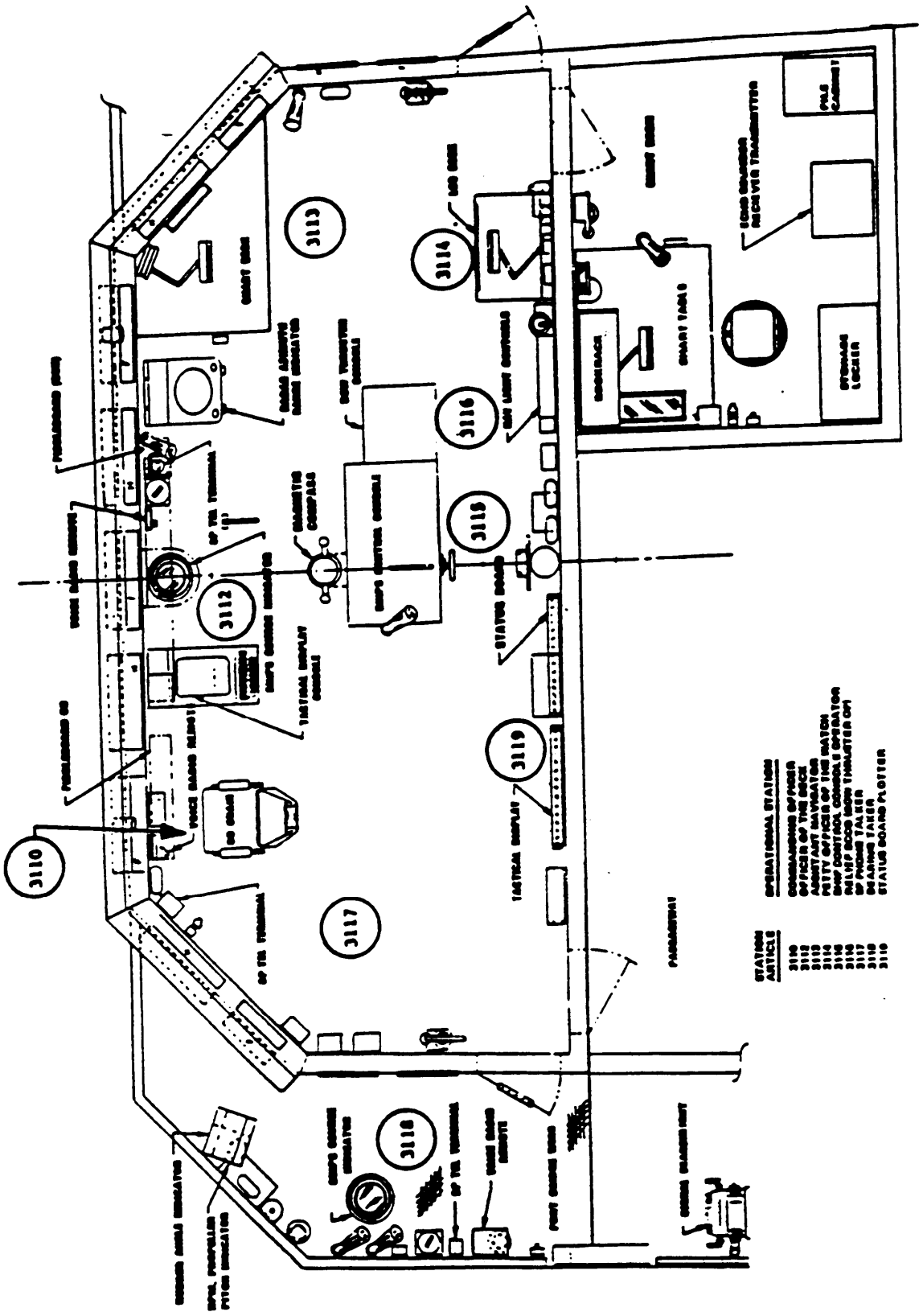


FIGURE 5-4. Space diagram.

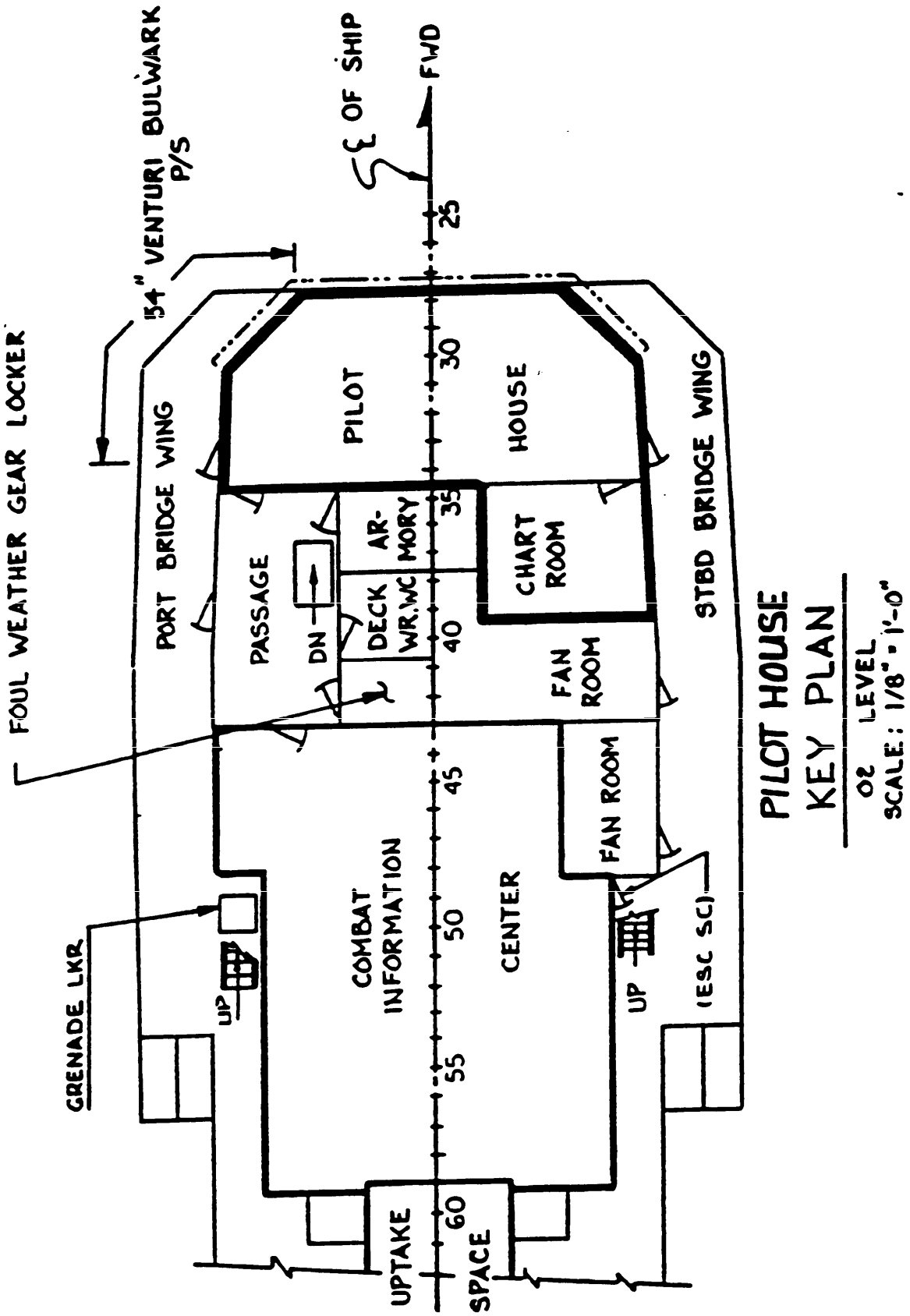


FIGURE 5-5. Key plan.

only be used when a network uses per or group calls in normal

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the comment number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of this form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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3. DOCUMENT TITLE
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4. NATURE OF CHANGE (identity paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

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A. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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A. NAME Technical Point of Contact (TPOC) MR. MICKEY ANDER, SEA 04TD3 ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial: DSN: TPOC: 703-602-8701 8-332-8701
c. ADDRESS (Include Zip Code) COMMANDER, NAVAL SEA SYSTEMS COMMAND ATTN: SEA 03R42 2531 JEFFERSON DAVIS HIGHWAY ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone 703-756-2340 DSN 289-2340